

Problem

If $f(t) = 10 + 8 \cos t + 4 \cos 3t + 2 \cos 5t + \dots$, the angular frequency of the 6th harmonic is

- (a) 12
- (b) 11
- (c) 9
- (d) 6
- (e) 1

Step-by-step solution

Step 1 of 1

$$f(t) = 10 + 8 \cos t + 4 \cos 3t + 2 \cos 5t + \dots$$

The angular frequency of 6_{th} harmonic is

$$= 2 \times 6 - 1$$

$$\boxed{= 11}$$